

DP Computer Science: A 2.3.4 Static and Dynamic Routing (HL Only)

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Differentiate between **static** and **dynamic** routing
 - Explain the **advantages and disadvantages** of each routing method
 - Analyse **factors influencing** the choice between static and dynamic routing
 - Evaluate the **suitability** of each method for different network scenarios
-

Key Vocabulary

Term	Definition
Routing	The process of forwarding data packets across networks from source to destination
Static routing	A routing method where routes are manually configured by a network administrator and do not change automatically
Dynamic routing	A routing method where routers automatically exchange information and update routing tables using protocols
Routing table	A database stored in a router that contains information about available routes and how to reach them
Routing protocol	A set of rules that routers use to communicate and share routing information (e.g., RIP, OSPF, BGP)
Convergence	The process by which routers in a network reach agreement on the best routes after a change
Redundancy	Having backup paths or components to ensure continued operation in case of failure

Term	Definition
Administrative overhead	The time and effort required to manage and maintain network configurations

ATL Skills

Skill	Description
Thinking	Critical thinking, analysing, comparing and contrasting, evaluating, decision-making
Communication	Explaining complex concepts, collaborating in groups, presenting information effectively
Research	Gathering and interpreting information from various sources
Self-Management	Time management, organisation, reflection

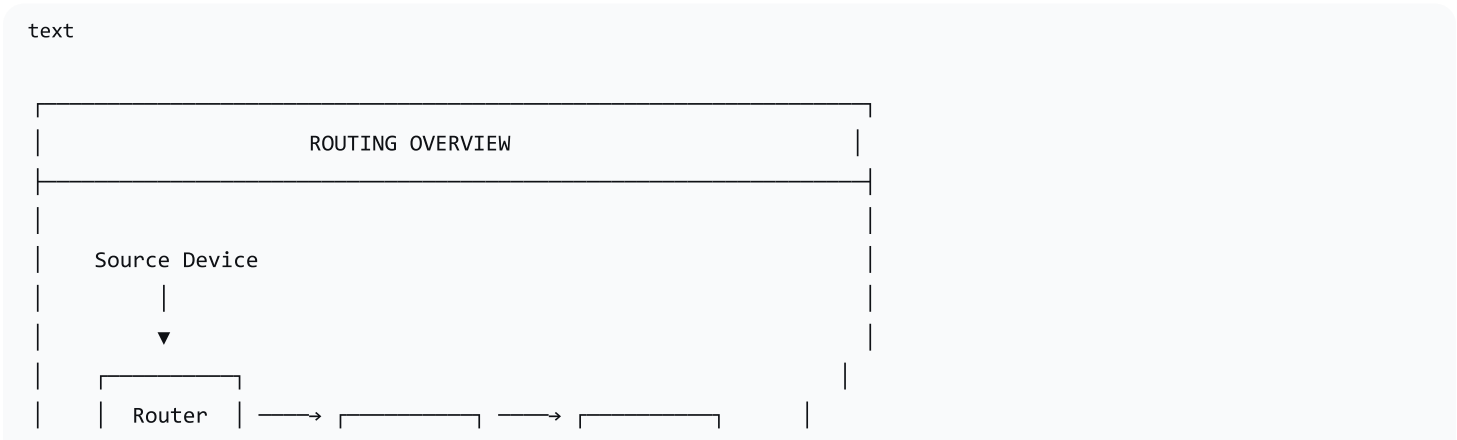
2. Introduction: The Routing Challenge

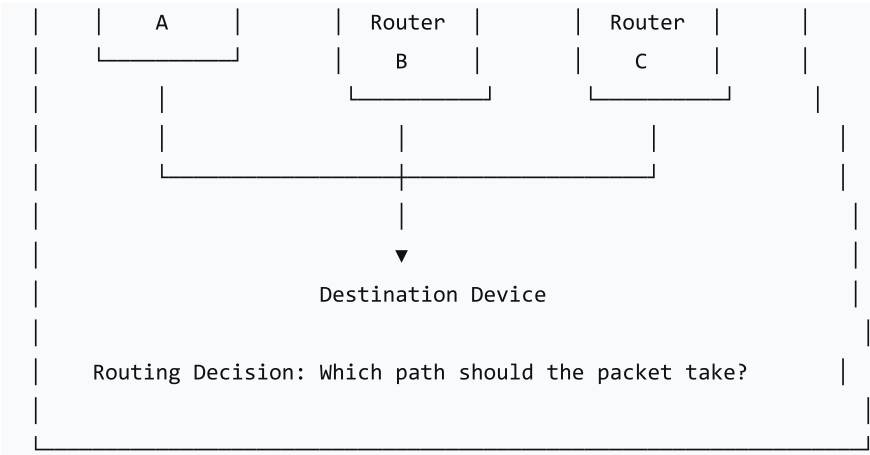
The Analogy

Imagine you're navigating through a city. You could follow a fixed route (static routing) or use a GPS that adapts to traffic and road closures (dynamic routing). Networks face a similar challenge when routing data.

What is Routing?

Routing is the process of forwarding data packets across networks from the source to the destination. Routers make decisions about the best path for each packet.





3. Static Routing

Definition

Static routing is a routing method where routes are manually configured by a network administrator and do not change automatically.

How It Works

Step	Description
1	Administrator manually enters routes into the routing table
2	Routes remain fixed unless manually changed
3	Router uses the static routing table to forward packets
4	If a route fails, traffic cannot reach that destination until the administrator updates the table

Advantages

Advantage	Description
Predictability	Routes are fixed and known; no unexpected changes
Security	No routing information is exchanged; less exposure to attacks
Efficiency	No bandwidth wasted on routing protocol updates

Advantage	Description
Simplicity	Easy to understand and configure for small networks
Control	Administrator has full control over routing decisions
Low Resource Use	No CPU/memory overhead from routing protocols

Disadvantages

Disadvantage	Description
Lack of Adaptability	Routes cannot automatically change in response to failures
High Maintenance	Manual updates required for any network change
Scalability Issues	Difficult to manage in large networks
No Fault Tolerance	No automatic failover; backup routes must be manually configured
Error-Prone	Manual configuration can lead to human errors
Time-Consuming	Adding or changing routes takes time

When to Use Static Routing

Scenario	Why Static Routing?
Small networks	Few routes; easy to manage manually
Simple topologies	Network structure rarely changes
Hub-and-spoke networks	Central site connects to fixed remote sites
Stable connections	Permanent dedicated links (e.g., point-to-point)
Security-sensitive networks	No routing information exchanged
Default routes	Simple "send everything here" configuration

Analogy: Static routing is like using a **paper map**—reliable for a fixed route, but if the road is closed, you need to manually find an alternative.

4. Dynamic Routing

Definition

Dynamic routing is a routing method where routers automatically exchange information and update routing tables using routing protocols.

How It Works

Step	Description
1	Routers use routing protocols to exchange information
2	Routing tables are automatically updated
3	Routers adapt to network changes (failures, congestion)
4	Convergence occurs when all routers agree on the best routes

Routing Protocols

Protocol	Type	Description
RIP (Routing Information Protocol)	Distance-vector	Simple, limited to 15 hops
OSPF (Open Shortest Path First)	Link-state	Fast convergence; suitable for large networks
BGP (Border Gateway Protocol)	Path-vector	Used on the internet backbone
EIGRP (Enhanced Interior Gateway Protocol)	Hybrid	Cisco proprietary; efficient and scalable

Advantages

Advantage	Description
Adaptability	Automatically adapts to network changes
Reduced Administration	No manual updates required for most changes
Scalability	Works well in large and complex networks

Advantage	Description
Fault Tolerance	Automatically reroutes around failures
Load Balancing	Can distribute traffic across multiple paths
Self-Healing	Recovers from failures without human intervention

Disadvantages

Disadvantage	Description
Complexity	More complex to configure and troubleshoot
Resource Usage	Consumes CPU, memory, and bandwidth
Security Risks	Routing information can be intercepted
Convergence Time	Takes time for routers to agree on routes after a change
Protocol Dependencies	Requires consistent protocols across routers
Vendor-Specific	Some protocols are proprietary

When to Use Dynamic Routing

Scenario	Why Dynamic Routing?
Large networks	Too many routes for manual management
Complex topologies	Many connections and potential paths
Networks with frequent changes	Growth, expansion, or reconfiguration
High-availability systems	Automatic failover required
Networks with redundancy	Multiple paths need to be managed automatically
Internet connections	BGP used for global routing

Analogy: Dynamic routing is like using **GPS navigation**—it adapts to traffic, road closures, and finds the best route automatically.

5. Comparison Table

Feature	Static Routing	Dynamic Routing
Configuration	Manual (administrator)	Automatic (routing protocols)
Adaptability	No — fixed routes	Yes — automatic updates
Scalability	Poor — not suitable for large networks	Excellent — works in large networks
Fault Tolerance	Low — no automatic failover	High — automatic rerouting
Security	High — no route info exchanged	Medium — route info is exchanged
Resource Usage	Low — no protocol overhead	High — uses CPU, memory, bandwidth
Administrative Overhead	High — manual updates required	Low — automatic updates
Complexity	Simple	Complex
Convergence Time	Instant (no changes)	Variable (routers exchange info)
Best For	Small, stable networks	Large, complex, changing networks
Example	Small office, point-to-point link	Enterprise, internet backbone

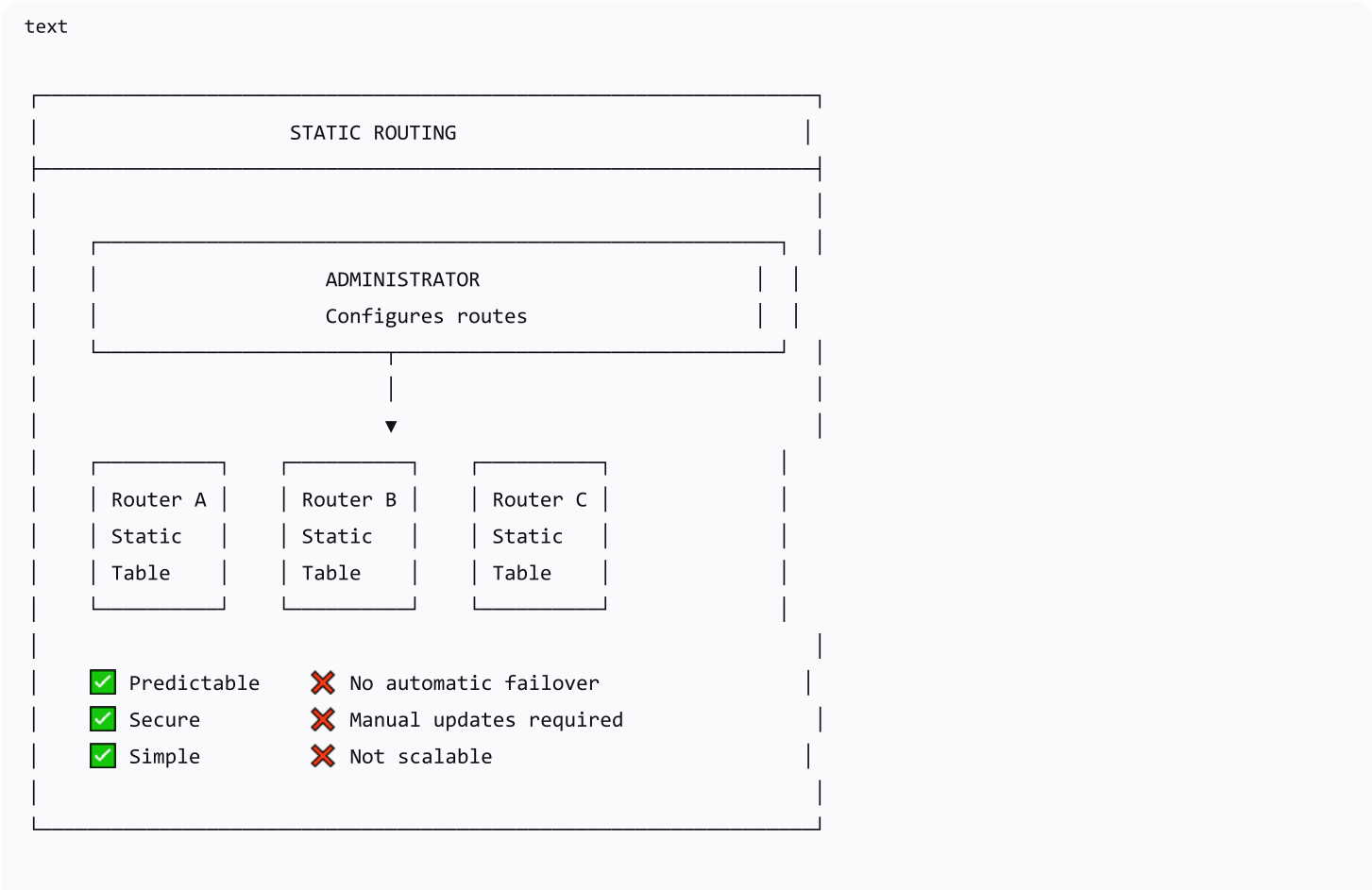
6. Factors Influencing Routing Choice

Factor	Static	Dynamic	Why?
Network Size	Small	Large	Static routes become unmanageable in large networks
Network Complexity	Simple	Complex	Dynamic routing handles complex topologies
Change Frequency	Low	High	Dynamic routing adapts to changes automatically

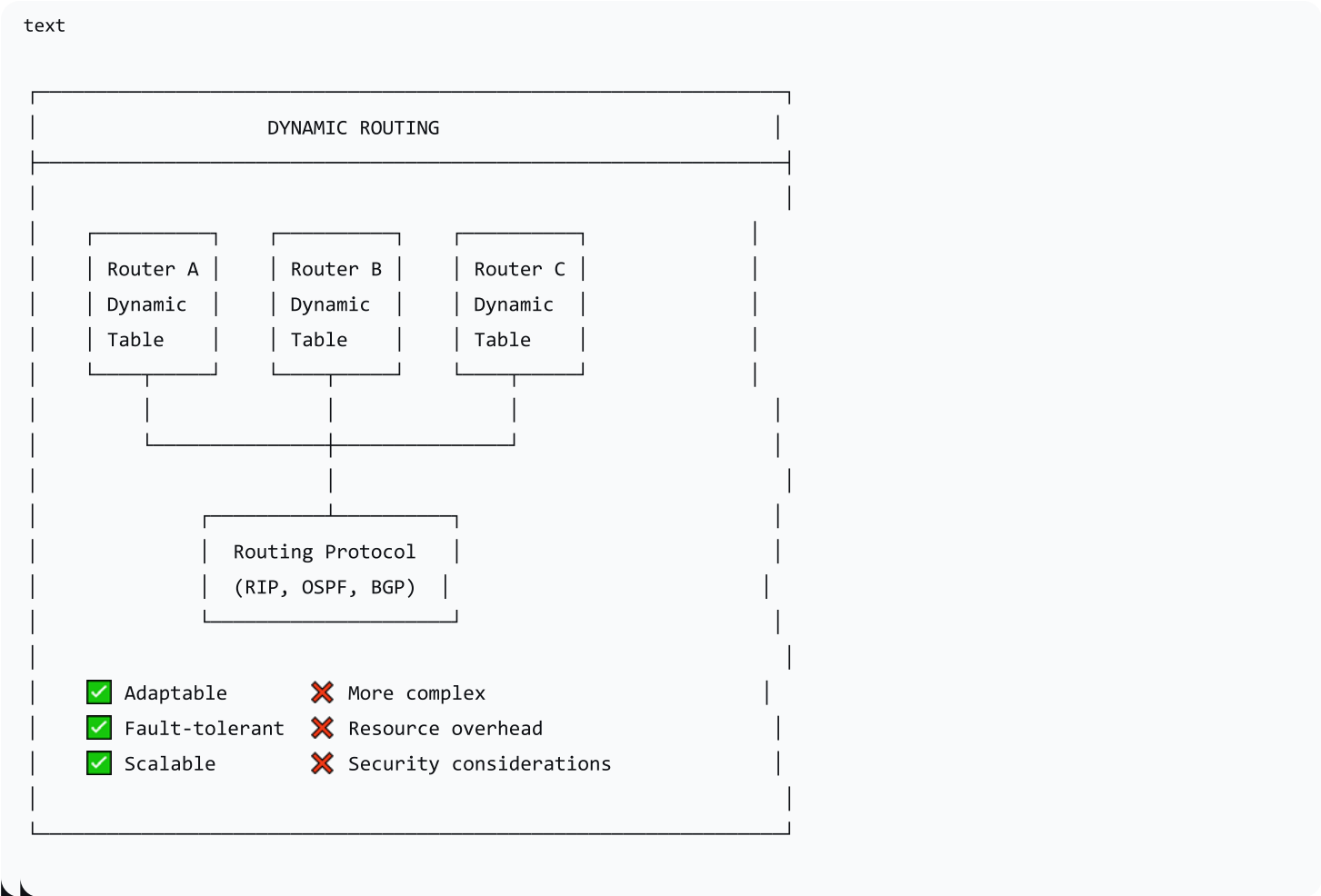
Factor	Static	Dynamic	Why?
Redundancy Requirements	Low	High	Dynamic routing provides automatic failover
Security Requirements	High	Medium	Static routing does not exchange routing information
Administrative Skills	Basic	Advanced	Dynamic routing requires knowledge of protocols
Budget	Low	High	Dynamic routing requires more expensive hardware
Reliability Requirements	Standard	High	Dynamic routing provides fault tolerance

7. Visual Comparison

Static Routing



Dynamic Routing



8. Routing Challenge Activity

Instructions

1. Divide class into groups
2. Each group analyses a network scenario
3. Decide whether to use static or dynamic routing (or a combination)
4. Justify the choice based on network size, complexity, and needs
5. Present solution and reasoning

Scenario 1: The Small Office Network

Description: A small office with 20 computers, a printer, and a file server. The network needs to be simple and easy to manage, with minimal resource usage.

Aspect	Consideration
Network Size	Very small (20 devices)
Complexity	Simple topology
Change Frequency	Rarely changes
Budget	Limited
Administrative Skills	Basic

Recommended: Static Routing

Why: Small network size, predictable traffic patterns, minimal administrative overhead. The administrator can manually configure routes between devices.

Scenario 2: The Branch Office Connection

Description: A company has main and branch offices in different locations. The branch office needs to connect to the main office to access shared resources securely.

Aspect	Consideration
Network Size	Two locations
Complexity	Moderate
Change Frequency	Low
Security	Important
Administrative Skills	Moderate

Recommended: Hybrid (Static + Dynamic)

Why: Static route for dedicated connection between offices (secure and predictable). Dynamic routing for internal traffic within each office (flexibility and adaptability).

Scenario 3: The High-Availability Web Server

Description: A company hosts a web server that must be highly available. The network should handle failures and maintain connectivity.

Aspect	Consideration
Network Size	Moderate
Complexity	Moderate
Redundancy	Critical
Fault Tolerance	Required
Uptime	Must be near 100%

Recommended: Dynamic Routing

Why: Dynamic routing automatically adapts to failures and redirects traffic through alternative paths. Ensures high availability and minimises downtime.

Scenario 4: The Growing Campus Network

Description: A university campus network with hundreds of devices and multiple departments. The network needs to scale for future growth.

Aspect	Consideration
Network Size	Large (hundreds/thousands of devices)
Complexity	Very complex
Change Frequency	High (growth, reconfiguration)
Scalability	Critical
Administrative Skills	Advanced

Recommended: Dynamic Routing

Why: Dynamic routing protocols automatically discover and adapt to network changes, making it easier to manage and scale as the network grows.

Scenario 5: The Multi-Homed Internet Connection

Description: A company has multiple internet connections from different ISPs for redundancy and performance.

Aspect	Consideration
Network Size	Moderate
Complexity	Moderate
Redundancy	Critical
Load Balancing	Required
Failover	Required

Recommended: Dynamic Routing (with multi-path support)

Why: Routing protocol can distribute traffic across available connections and automatically switch to a backup if one fails. Ensures continuous internet access and efficient load balancing.

9. Lesson Sequence

Lesson Plan

Phase	Time	Activity
Introduction	5 min	"Navigating a city" analogy; introduce static and dynamic routing
Static Routing	15 min	Manual configuration; advantages/disadvantages; suitable scenarios
Dynamic Routing	15 min	Routing protocols; advantages/disadvantages; suitable scenarios
Routing Challenge	20 min	Group activity—analyse scenarios, decide routing method, justify choice
Reflection	5 min	Recap; class discussion; assign homework

10. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
Routing Challenge	Assess ability to analyse scenarios and justify routing choices
Quiz	Multiple-choice quiz on eBook page A2.3.4
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. What is the difference between static and dynamic routing?
2. Give one advantage of static routing.
3. Give one advantage of dynamic routing.
4. Which routing method would you use for a small office network? Why?

11. Differentiation

Level	Strategy
Support	Provide visual aids, concise definitions, and simplified scenarios
Challenge	Encourage research on specific routing protocols (RIP, OSPF, BGP) and their applications

12. Resources

- Presentation (Slide Deck)
- eBook (A2.3.4 with quiz)
- Activity: Routing Challenge scenarios
- Quizlet (A2.3.4 vocabulary flashcards)

13. Summary: Key Takeaways

Method	Key Feature	Best For
Static Routing	Manually configured; fixed routes	Small, stable networks; point-to-point links

Method	Key Feature	Best For
Dynamic Routing	Automatically updated; adapts to changes	Large, complex, growing networks
Hybrid	Combination of both	Branches with dedicated connections; internal flexibility

Choosing Between Static and Dynamic Routing:

Factor	Static	Dynamic
Network Size	Small	Large
Complexity	Simple	Complex
Change Frequency	Low	High
Redundancy	Low	High
Security	High	Medium
Administrative Skills	Basic	Advanced

"The choice between static and dynamic routing depends on the specific needs of the network—size, complexity, change frequency, and redundancy requirements all play a role in the decision."